



How frictional response during solute solvation controls solute rotation in naturally abundant deep eutectic solvent (NADES)? A case study with amino acid derivative containing DES

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ARTICLE INFO

Article history:

Received 1 October 2019

Received in revised form 27 December 2019

Accepted 2 January 2020

Available online 9 January 2020

Keywords:

Naturally abundant deep eutectic solvent

Temperature dependent measurements

Stokes shift dynamics

Time-resolved fluorescence anisotropy

Viscosity decoupling

ABSTRACT

We explored frictional response of a room temperature naturally abundant deep eutectic solvent made of betaine, urea and water ([Bet:Ure:Wat (11.7:12:1 weight ratio)]) via temperature dependent measurements of solute-centered dynamics, and characterized the coupling between the relaxation rates and medium friction. Time-resolved fluorescence studies with this medium at different temperatures revealed fractional viscosity dependencies for solute rotation and solvation rates, suggesting strong temporal heterogeneity in the medium. Intriguingly, such a substantial viscosity decoupling was found to occur at a temperature ~ 90 – 130 K above the measured glass transition temperature ($T_g \sim 218$ K) of this amino acid derivative containing DES. Estimated activation energies from dynamic fluorescence anisotropies of hydrophobic and hydrophilic solutes were nearly half (~ 30 kJ mol⁻¹) of that estimated from temperature dependent viscosity measurements (~ 57 kJ mol⁻¹). Both rotational and solvation energy relaxations for these solutes were found strongly non-exponential yet the friction that controlled solute's rotation in this biodegradable deep eutectic solvent did not fully map the resistance that medium particles experienced while solvating the solute. This is new, and contrasting to results found for several complex systems that include ionic deep eutectics, ionic liquids, and confined water. This observation should be reexamined via more sophisticated measurements. Temperature dependent densities and refractive indices were measured to understand the dipolar solvation response in this medium, particularly the correctness of the missing components estimated from a well-known approximate method developed for conventional room temperature polar solvents.

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1. Introduction

Demand for environment friendly and less toxic solvent leads to the invention of new classes of solvents such as room temperature supercritical fluids (RTSFs) [1–3], gas expanded liquids (GXLs) [4–6], room temperature ionic liquids (RTILs) [7–10] and deep eutectic solvents (DESSs) [11–16]. Deep eutectic solvents (DESSs) provide a new generation potential alternatives to the conventional molecular solvents for large scale industrial applications because of exquisite solvent properties. Plentiful natural and biodegradable chemicals along with a large variety of possible combinations, easy preparation, low cost, easy transportation, minimal vapor pressure and other physicochemical features have rendered DESs not only a system for wide-spread applications but motivated also studies of basic science aspects probing the nature of frictional coupling of reactive modes and the subsequent medium control. Proper selection of eco-friendly and biodegradable components can provide DESs which may be termed as green solvents [13,17–22].

Desirable properties of DESs can be achieved by appropriate selection of its components and composition [23–25]. Thus, the selection of the components for DES preparations is vital, and one prefers here the naturally abundant compounds, such as, amino acids, urea, sugars etc. to preserve greenness and eco-sustainability of the resultant DESs [19,22,26–30]. DESs prepared from these natural materials are known as naturally abundant deep eutectic solvents (NADESs) and are already in use in various interdisciplinary areas [20,28,31–36].

In this work we have taken betaine, urea and water as components for preparing a NADES, and the subsequent study (see scheme1). Betaine is a trimethyl derivative of glycine (amino acid) present in bacteria, invertebrates, plants, various foods and mammals [37–41]. Moreover, non-toxic nature encourages selection of betaine as one of the components of DES [42]. Betaine is used as an osmolyte [43], a methyl group source for different biochemical reactions [44], and in treatment of several diseases [45–47]. Another component of the present DES is urea, the most common nitrogenous fertilizers which has deep impact on protein stability and functionality [48–50]. Interestingly, betaine can be served as an osmolyte in the preservation of protein against the extreme physical conditions [51], while urea acts as a protein denaturant [49,50]. In

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such a scenario the stability of protein depends on a novel interplay of opposing interactions guided by the ratio of the osmolyte and the denaturant [52,53]. Therefore, betaine, urea and water based DES would be an important system for understanding interspecies interactions and the mixture dynamics. Physical properties of betaine or urea based DESs were studied before and used for various purposes [11,54–59]. Betain-urea based deep eutectics have been employed for stabilization of proteins [58,60], extraction of bioactive compounds [61], for applications to suit biomedical needs [62], etc. For preservation and stabilization of DNA, one needs to understand how DNA molecules interact with these novel solvent systems as DNA-medium interaction play a critical role [63]. Exploration of interaction and dynamics of these DESs is of utmost importance for understanding several biochemical processes that involve water-insoluble but lipid-soluble metabolites because these lipid-soluble metabolites get dissolved rather easily in such DESs [28,30].

It is now understood that property-driven applications of any class of solvents including DESs demand comprehensive knowledge of the inherent interactions and dynamics. These in several DESs have been explored, like in conventional media, via a variety of experimental techniques such as neutron diffraction [64,65], pulsed field gradient NMR [66,67], dielectric relaxation [16,25,68], fs-RIKES [69], 2D-IR [70,71], and time-resolved fluorescence [15,24,68,72], and also by performing computer simulations [73–79]. Here, we have made an attempt to investigate and understand interactions and relaxation dynamics of betaine, urea and water ([Bet+Ure+Wat]) based DES employing steady state and time-resolved fluorescence measurements. The emphasis here is on the frictional response generated by this biodegradable NADES, and how this frictional response interconnects two non-reactive solute-centered dynamical information supplied by the dynamic fluorescence Stokes shift and the time-resolved fluorescence anisotropy measurements. During fluorescence depolarization via orientational relaxation, the excited solute interacts with the neighboring solvent particles and overcomes the resultant resistance; therefore the dynamic anisotropy measurements provide a qualitative description of the time-dependent friction acting on the rotating excited solute. The same frictional profile dictates the time-dependent solvation energy relaxation of that excited solute probed by the dynamic fluorescence Stokes shift experiments. Therefore, the solute solvation and rotation is connected at the dynamic friction level, and one expects a proportionality relationship between the average rates (or timescales) of solute solvation and rotation if no other factors influence these two solute-centered non-reactive relaxation rates. The information regarding the frictional profile is important to understand the solvent dynamic effects on a chemical reaction and is therefore intimately connected to liquid solvent engineering for smarter applications in industrial and technology sectors. In the present work we have explored the interrelationship between solute solvation and rotation by employing two solutes of different chemical nature: a hydrophobic probe, coumarin 153 (C153), and a hydrophilic one, coumarin 343 (C343). This is done to investigate the inhomogeneity aspect of this ternary mixture which possesses meta-stability because of its preparation protocol.

2. Experimental details

2.1. Sample preparation

Betaine ($\geq 99.0\%$, Sigma-Aldrich, melting point $T_m \sim 574$ K) and Urea ($\geq 98\%$, Sigma-Aldrich, $T_m \sim 405$ – 408 K) were vacuum-dried (~ 300 K) overnight before use; Millipore water was used for sample preparation. Laser grade C153, C343, and trans-2-[4-(dimethylamino)styryl] benzothiazole (DMASBT) – all from Sigma-Aldrich – were used as received. Betaine, urea and water [Bet:Ure:Wat (11.7:12:1; weight ratio)] were taken in a screw-capped glass vial and heated at ~ 343 K with constant stirring at ~ 600 rpm in a hot oil bath for ~ 2 h. A colorless transparent liquid (that is, the DES) was formed and allowed to cool

gradually to room temperature (~ 298 K). For solubility checking, required amount of three components were taken in a container and kept under constant stirring at 600 rpm at 298 K. This mixture became opaque ('B' of Fig. 1). This opaque mixture never transformed into a transparent solution at 298 K even after continuous stirring for a considerable period of time. A repeat of this process with temperature ~ 10 K and ~ 20 K above 298 K also did not produce any transparent solution. Dissolution of the other two components by water was not possible because water was present approximately in 1:24 (by weight) ratio. The resultant DES prepared following the 'appropriate' method was a transparent liquid at that temperature ('A' of Fig. 1; $T_f \sim 279 \pm 2$ K). All these observations suggest that the present NADES is not a -simple-three component mixture but a deep eutectic solvent only. Note that composition of betaine, urea and water for the DES prepared here are different from the existing one [33] and thus the present one may be termed as a new deep eutectic solvent. DES, prepared via following the method described above, was subsequently transferred into a preheated quartz cuvette (path length 1 cm), which was already loaded with the grains of fluorescent probes (one at a time) mentioned above. Before measurements, sufficient time was allowed for the sample containing cuvette (placed into the designated sample chamber) for complete dissolution of the fluorescent solute and thermal equilibration (with uncertainty ± 1 K). Note that external fluorescent probe (C153, C343 and DMASBT for chemical structures see Scheme 1) concentration was maintained at $\leq 10^{-5}$ M in all optical measurements.

2.2. Differential scanning calorimetric (DSC) measurements

Glass transition temperature (T_g) of the prepared DES was measured by a differential scanning calorimeter (DSC, TA Instrument Q2000) [72]. The calorimetric trace is shown in Fig. 2 and the indicated glass transition temperature is ~ 218 K. All measurements reported here were performed well above (~ 90 – 130 K) the T_g (~ 218 K) as the temperature range scanned was, $303 \leq T/K \leq 343$.

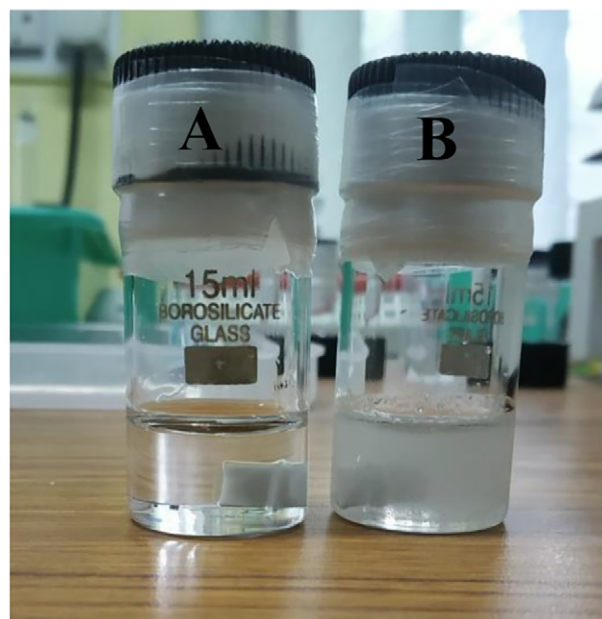
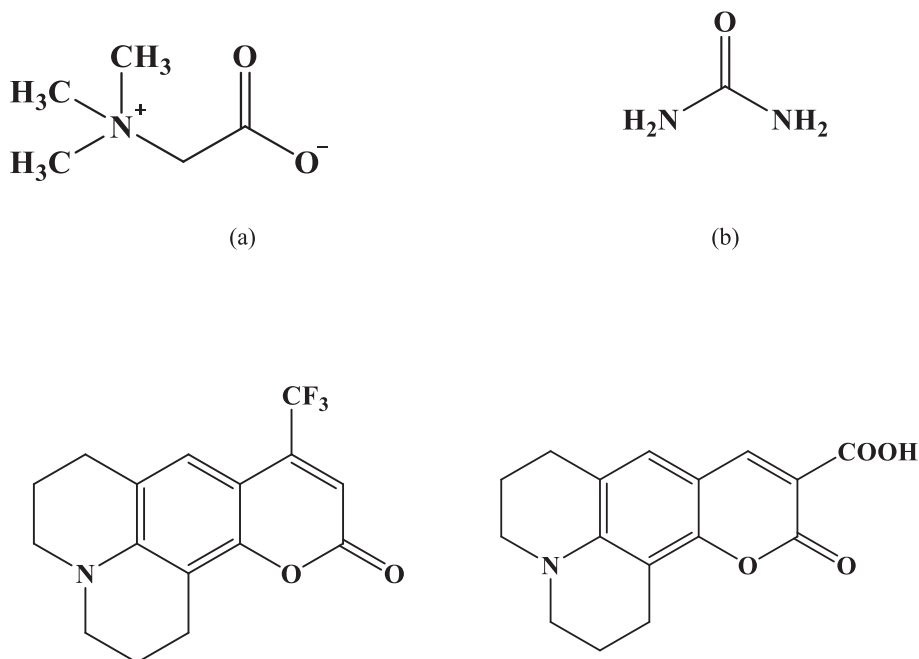


Fig. 1. A and B are the two [Bet+Ure+Wat (11.7:12:1; weight ratio)] mixtures. 'A' was prepared by mixing all the constituents and heating at ~ 344 K as described in Appendix 1: Sample preparation. 'B' was obtained by preparing the mixture, shaking occasionally, and allowing it to be at 298 K. This photo was taken when both the systems were kept at room temperature (~ 298 K). Mixture, B, is opaque and semi-solid whereas A provides a transparent colorless liquid, forming DES.



Scheme 1. Chemical structures of (a) betaine, (b) urea, (c) coumarin 153, and (d) coumarin 343

2.3. Refractive index, density and viscosity coefficient measurements

A refractometer (RUDOLPH, J357), an automated density-cum-sound analyzer (Anton Paar, DSA5000) and a micro viscometer (AMVn, Anton Paar), were employed respectively for refractive index (n_D), density (ρ) and viscosity coefficient (η) measurements [14,72]. Temperature dependent n_D , ρ and η are summarized in Table 1. Note that η of this DES varies approximately by an order of magnitude within the experimental temperature range ($303 \leq T/K \leq 343$).

2.4. Steady-state absorption and fluorescence measurements

UV-Visible spectrophotometer (UV-2600, Shimadzu) and fluorimeter (Fluorolog, Jobin-Yvon, Horiba) connected with Peltier-temperature controller were used for steady state absorption and fluorescence emission measurements, respectively. Prior to analysis, solvent blanks were subtracted from the probe spectra and transformed appropriately to the

frequency domain for frequency determination and other spectral properties [21,68].

2.5. Time-resolved fluorescence measurements

Time-correlated single photon counting (LifeSpec-ps, Edinburgh Instruments, U. K.) technique along with 409 nm (LED) excitation was employed for time-resolved fluorescence measurements [80,81]. The full width at half-maximum (FWHM) of the instrument response function was ~85 ps. Subsequently, emission decays were collected at the peak wavelength of the steady state emission spectra in two different polarizations—parallel ($I_{para}(t)$), and perpendicular ($I_{perp}(t)$) with respect to the polarization of the exciting light [81–84]. The dynamic fluorescence anisotropy, $r(t)$, was obtained from the collected intensity decays as follows

$$r(t) = \frac{I_{para}(t) - G I_{perp}(t)}{I_{para}}(t) + 2G I_{perp}(t) \quad (1)$$

The average rotational times, $\langle \tau_r \rangle$, were then calculated analytically from the bi-exponential fits to the $r(t)$ decays, $\langle \tau_r \rangle = \int_0^\infty dt [r(t)/r_0] = \int_0^\infty dt \sum_{i=1}^2 a_i \exp(-t/\tau_i) = a_1 \tau_1 + a_2 \tau_2$ with $a_1 + a_2 = 1$, $r_0 = 0.376$ for C153 [82] and 0.35 for C343 [85]. G , the geometric factor, was obtained by tail matching the intensity decays $I_{para}(t)$ and $I_{perp}(t)$, and found to be 1.4 ± 0.3 for C343 and 1.6 ± 0.2 for C153. We would like to mention here that the values of r_0 for both the solutes were fixed and not considered as free parameter during fits to temperature dependent $r(t)$ decays because our principal focus was to qualitatively understand the impact of medium friction on solute's rotational dynamics, and the way the medium friction interconnects the solvation and rotation of the same solute in that medium. A dependence of this frictional coupling on the chemical nature of the probe solute was another important aspect of this investigation.

For the dynamic Stokes shift measurements 14–16 fluorescent transients (magic angle) were collected at equally spaced wavelengths across the steady state emission spectrum of C343 (C153) dissolved in [Bet+Ure+Wat] DES. Subsequently, time-resolved emission spectra (TRES) were reconstructed following an established protocol [86,87]. The time dependent progress of solvation of the excited dipolar solute

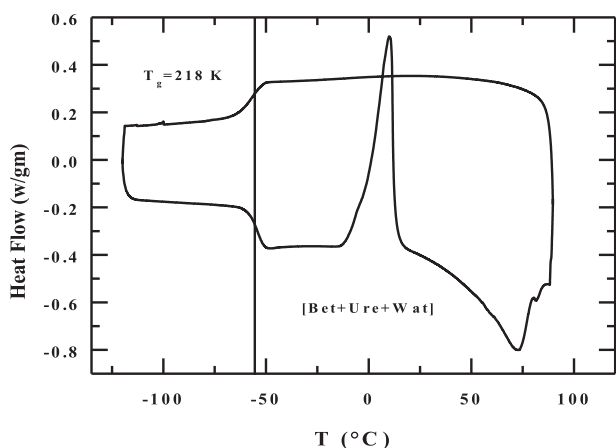


Fig. 2. Differential scanning calorimetric (DSC) trace for [Bet+Ure+Wat] DES. The glass transition temperature (T_g) is indicated.

Table 1
Temperature dependence of density, viscosity and refractive index of [Bet+Ure+Wat] DES.

T(K)	Viscosity ^a (cP)	Refractive ^b index	Density ^b (g/cm ³)
303		1.49656	1.219047
	431	1.49672	1.216162
308			
	281	1.49689	1.213264
313			
	192	1.49691	1.210346
318			
	136	1.49696	1.207398
323			
	99	1.49702	1.204443
328			
	74	1.49721	1.201593
333			
	56	1.49745	1.198734
338			
	45	1.49757	1.195861
343			

^a Uncertainty ± 10%.
^b The uncertainty is ±5% of the last three digits of the numbers reported here.

was then monitored via the normalized solvation response function [86],

$$S(t) = \frac{v(t) - v(\infty)}{v(0) - v(\infty)}, \tag{2}$$

where $v(x)$ representing the fluorescence frequency at time 0, t and ∞ , respectively. The average solvation times, $\langle \tau_s \rangle$, were estimated via time-integrating the bi-exponential fits to the $S(t)$ decays: $\langle \tau_s \rangle = \int_0^\infty dt S(t) = \int_0^\infty dt \sum_{i=1}^2 a_i \exp[-t/\tau_i] = \sum_{i=1}^2 a_i \tau_i$, with $\sum_{i=1}^2 a_i = 1$. Magic angle (54.7°) intensity decays at the peak wavelength of the steady state

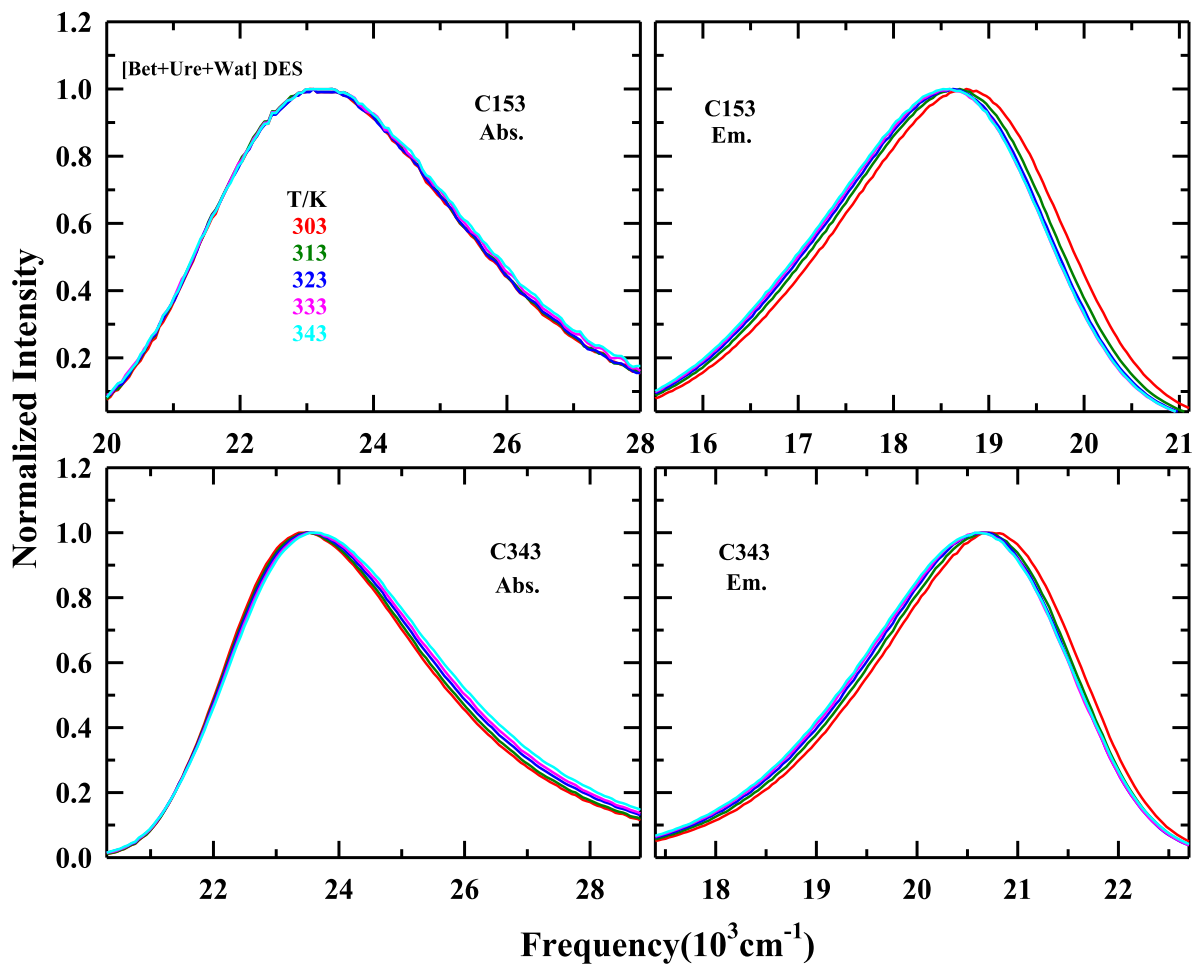


Fig. 3. Temperature dependent absorption (left panels) and emission (right panels) spectra of C153 and C343 in [Bet+Ure+Wat] DES. All representations are color coded.

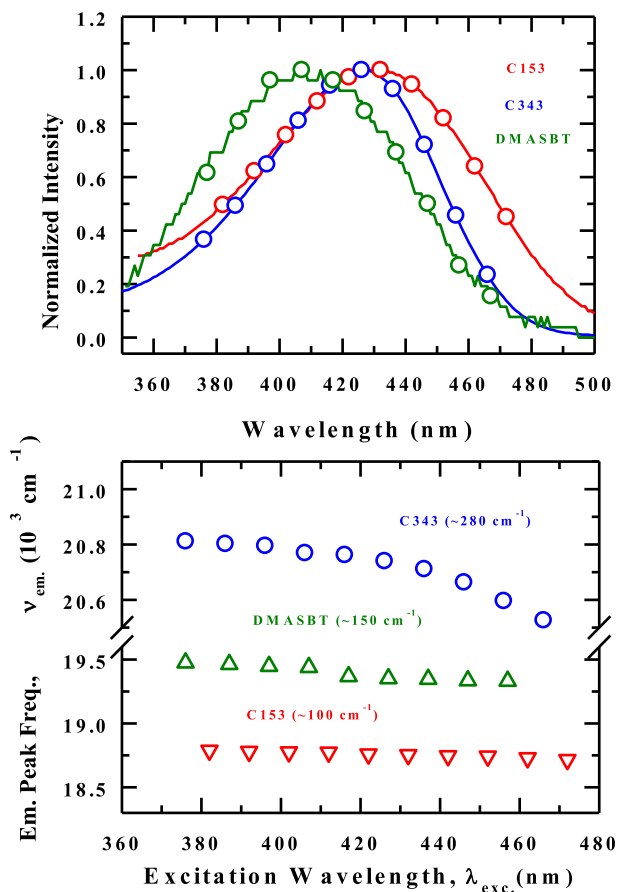


Fig. 4. Representative absorption spectra of C153, C343 and DMASBT in [Bet+Ure+Wat] DES at 303 K (upper panel). The circles on the absorption spectra indicate the wavelengths selected for excitation wavelength dependence studies. Excitation wavelength ($\lambda_{exc.}$) dependence of fluorescence emission peak frequencies (ν_{em}) for C153, C343, and DMASBT in [Bet+Ure+Wat] DES at 303 K (lower panel). All the representations are color coded.

emission spectra of C153 and C343 dissolved in [Bet+Ure+Wat] DES were collected in the temperature range $303 \leq T/K \leq 343$ for investigating, as before [88], the temperature dependence of the fluorescence lifetimes for these probes dissolved in this DES.

3. Results and discussion

3.1. Characterization of NADES

NADES prepared from betaine, urea and water [Bet:Ure:Wat (11.7:12:1; weight ratio)] is a transparent liquid at much below the room temperature ($T_f \sim 279 \pm 2K$). This new NADES has been characterized by measuring several important quantities, such as, T_g ($\sim 218 K$), refractive index, density, and viscosity coefficient (see Table 1).

3.2. Steady state absorption and emission spectral features: hydrophobic C153 and hydrophilic C343 probes

Temperature ($303 \leq T/K \leq 343$) dependent steady state absorption and emission spectra of C153 and C343 in [Bet+Ure+Wat] DES are shown in Fig. 3. The spectral features of C153 and C343 are similar and nearly insensitive to the temperature, though one expects that C153 and C343, owing to their different chemical nature, would locate in different environments and reflect different temperature dependencies. This observation therefore suggests that the solute-solvent interactions remain nearly unchanged for both C153 and C343 within the

temperature range $303 \leq T/K \leq 343$. In DESs this insensitivity to temperature of steady state spectral behavior of C153 has already been observed [21,68]. This is somewhat counter-intuitive because rise in temperature is expected to reduce the static dielectric constant (and hence polarity) of the medium [89–91]. Excitation wavelength ($\lambda_{exc.}$) dependence of the fluorescence emission of these solutes in this DES was then carried out to investigate the solution spatial heterogeneity. Excitation wavelengths chosen for this study are shown in the upper panel of Fig. 4 while lower panel of this figure presents the $\lambda_{exc.}$ dependence of fluorescence emission peak frequencies (ν_{em}) of C153 and C343. $\lambda_{exc.}$ -induced emission shift, $\Delta\nu_{em} = \nu_{em}^{peak}(\lambda_{blue, exc.}) - \nu_{em}^{peak}(\lambda_{red, exc.})$, obtained for C153 and C343, is within ~ 100 – 300 cm^{-1} which suggests that the present DES is mildly heterogeneous within the lifetime (see Table S1, Supporting material) of these probes. In addition, $\lambda_{exc.}$ dependence of ν_{em} for a short lifetime probe, DMASBT ($\langle\tau_{life}\rangle \sim 750 \text{ ps}$ in this DES) was monitored, and similar shift in the ν_{em} found (see Fig. 4). All these strongly indicate that the solution structure of the present DES is mildly heterogeneous within the lifetimes (~ 1 – 5 ns) of these fluorescent probes. Note this ‘mildly heterogeneous’ or near-homogeneous solution structure arises from an interpretation of the $\lambda_{exc.}$ -induced shift in ν_{em} , which (the emission shift) depends strongly on a competition between $\langle\tau_{life}\rangle$ of a dissolved probe and the timescales in which different solvation environments surrounding the probe molecules inter-convert. Therefore, too fast environmental fluctuations would remain undetectable in the steady state measurements with probes that have relatively longer lifetimes, making the solution structure appearing as ‘homogeneous’, although significant temporal heterogeneity (occurring at a timescale faster than the shortest $\langle\tau_{life}\rangle$)

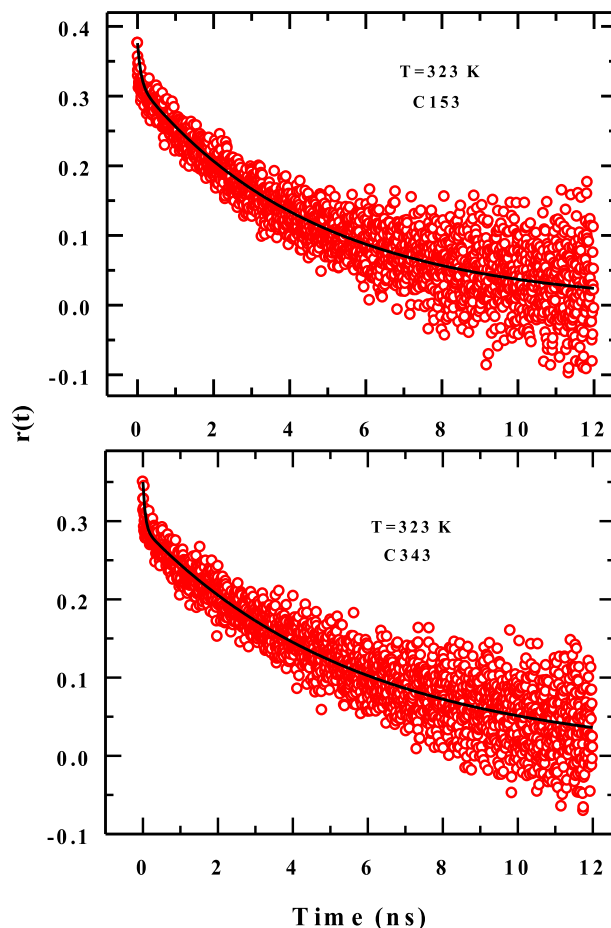


Fig. 5. Representative time-resolved fluorescence anisotropy ($r(t)$) decays of C153 (upper panel) and C343 (lower panel) in [Bet+Ure+Wat] DES at 323 K. Lines going through data denote bi-exponential fits.

involved here) may still dictate the motional features of the medium particles.

Temperature dependent ($303 \leq T/K \leq 343$) average fluorescence life-times ($\langle\tau_{life}\rangle$) for C153 and C343 in this DES are summarized in Table S1. Notice $\langle\tau_{life}\rangle$ for C343 in this DES is insensitive to temperature, whereas that for C153 registers a decrease of $\sim 30\%$. This may suggest that the non-radiative rates (k_{nr}) for these probes sense differently the temperature impact on their respective solvation environments.

3.3. Solute rotation: coupling to medium friction

Time-resolved fluorescence intensity decays, $I_{para}(t)$ and $I_{perp}(t)$, of C153 (upper panel) and C343 (lower panel) dissolved in [Bet+Ure+Wat] DES are shown in Fig. S2 (Supporting material). Representative rotational anisotropy decays, $r(t)$, at 323 K for C153 (upper panel) and C343 (lower panel) in this DES, and the corresponding bi-exponential fits are displayed in Fig. 5. Fit parameters are summarized in Table 2. Parameters from bi-exponential fits to $r(t)$ decays at other temperatures in the range, $303 \leq T/K \leq 343$, are also provided in Table 2. Note these bi-modal decays are characterized by a minor ($\sim 20\%$) component with time constant ~ 100 ps, followed by a dominating, slower component with a time constant of a few nanoseconds. As increase in temperature leads to decrease in solution viscosity (see Table 1), the friction experienced by a rotating solute is expected to decrease. Therefore, the decrease in the average solute rotation times, $\langle\tau_r\rangle$, with temperature follows a trend same as that shown by the DES viscosity (see Table 2). The coupling between solute rotation and solution friction is explored next by showing $\langle\tau_r\rangle$ as a function of temperature-scaled viscosity, η/T ,

in a log-log plot in Fig. 6 with data for C153 in the upper panel and for C343 in the lower panel. A fractional viscosity dependence of the solute rotation times of the type, $\langle\tau_r\rangle \propto (\eta/T)^p$ with $p \approx 0.5$ for both C153 and C343, suggests significant temporal heterogeneity in the rotational relaxation rates for these solutes in this NADES. Such a strong fractional viscosity dependence has already been observed for solute rotation in ionic acetamide DESs [14,92]. Interestingly, both C153 and C343, irrespective of their different chemical nature, exhibit similar decoupling from medium viscosity while executing rotational diffusion in this DES.

In the same figure (Fig. 6), coupling to hydrodynamic frictions for these solutes in the limit of stick and slip boundary conditions are shown (dashed lines) after estimating the rotation times using the formula [23,93,94], $\tau_r = \eta V f C / k_B T$. In these calculations, volume $V^{C153} = 246 \text{ \AA}^3$ ($V^{C343} = 243 \text{ \AA}^3$), shape factor $f^{C153} = 1.71$ ($f^{C343} = 1.99$), $C_{stick}^{solute} = 1$, and $C_{slip}^{C153} = 0.24$ [82] ($C_{slip}^{C343} = 0.18$ [95]) were used. Note the measured $\langle\tau_r\rangle$ for C153 and C343 fall between stick and slip predictions at higher temperatures (or at lower η/T values), but become sub-slip at lower temperatures (higher η/T). This deviation from hydrodynamic predictions confirms partial decoupling of the solute rotation from the frictional resistance arising from the medium viscosity. This kind of decoupling may also originate from non-hydrodynamic angular moves, such as, orientation jumps and persistent inertia-driven motion [96–98].

Next, estimated activation energies (E_a) associated with solute (C153/C343) rotation and viscosity of the medium are shown in Fig. 7. E_a associated with the solute rotation is expected to be similar to that from viscosity if solute rotation is regulated by the friction generated exclusively by the medium viscosity. In [Bet+Ure+Wat] DES,

Table 2
Bi-exponential fit parameters for $r(t)$ of C153 and C343 in [Bet+Ure+Wat] DES at various temperatures.

$T(K)$	a_1 (%)	τ_1 (ps)	a_2 (%)	τ_2 (ps)	$\langle\tau_r\rangle^a$ (ps)
C153 in [Bet+Ure+Wat] DES					
303					
	20	84	80	6812	5466
308	18	70	82	5731	4712
313	16	89	84	5038	4246
318	15	92	85	4651	3967
323	11	118	89	3742	3343
328	14	98	86	2866	2478
333	12	108	88	2298	2035
338	14	114	86	1707	1484
343					
C343 in [Bet+Ure+Wat] DES					
303					
	21	62	79	8598	6805
308	20	54	80	7760	6219
313	14	57	86	6464	5567
318	17	62	83	5753	4786
323	9	65	91	4602	4194
328	13	80	87	3837	3349
333	14	72	86	3042	2626
338	10	61	90	2301	2077
343					

^a Average rotational time, $\langle\tau_r\rangle$, can be reproduced within $\pm 10\%$ uncertainty.

temperature dependent measurements of solution viscosity provide an activation energy, E_a^{η} ~56.7 kJ mol⁻¹, which is ~2 times greater than that obtained from temperature dependent C153/C343 rotation times (E_a^{C153} ~31.4 kJ mol⁻¹ and E_a^{C343} ~29.7 kJ mol⁻¹). This is a direct reflection of the fractional viscosity dependence of solute rotation times shown already in Fig. 6.

3.4. Solvation dynamics

Fig. S3 presents fluorescence transients of C343 in [Bet+Ure+Wat] DES collected at the short (blue) emission wavelength (at 450 nm) and long (red) wavelengths (at 590 nm) along with the tri-exponential fits. Note here that our similar measurements using C153 could not detect any Stokes shift dynamics, presumably because of the location of the probe where rearrangement of the surrounding medium was too fast to be detected in the present experimental set-up. Parameters of the tri-exponential fits to the above C343 intensity decays are provided in the inset. Only decay at the lower wavelength (blue), and rise followed by decay at the higher wavelength (red) is a hallmark of Stokes shift dynamics [23,99]. Fig. S3 displays this feature for C343 in [Bet+Ure+Wat] DES at 303 K.

Time-resolved emission spectra (TRES) along with the steady state emission spectrum are shown in Fig. 8 (upper panel). Blue shift in the steady state spectrum relative to TRES at $t = \infty$ indicates incomplete equilibration of solvent environments surrounding the excited C343 in [Bet+Ure+Wat] DES during the steady state fluorescence emission. This has also been suggested by a moderate amount (~300 cm⁻¹) of λ_{exc} -induced shift of ν_{em} shown in Fig. 4. Magnitudes of temperature dependent dynamic Stokes shift for C343 in this DES are ~700–800 cm⁻¹ with estimated missing component of ~70–75% (calculated using the Fee-Maroncelli method [100]) of the total dynamics. This means the broad temporal resolution (~85 ps) employed in the present measurements could detect only the slower portion of the dynamics, leaving out completely the sub-picosecond and sub-50 ps relaxations. These relaxation timescales are quite feasible for this DES because it contains small dipolar molecules (urea and water), and all the constituents are capable of participating in H-bonding. Collective low frequency motions of H-bonded species are known to generate sub-picosecond solvation response [101,102], while orientational rearrangements of DES particles can contribute to the sub-50 ps dynamics. Temperature dependent hydrodynamic rotation times of water, urea and betaine molecules in this DES, summarized in Table S4 (Supporting material), support this conjecture. Missing percentage increases with increasing temperature (see Table 3). We would like to mention here that a large density difference between non-polar reference solvent (hexane, ρ ~0.65 g/cm³) [103] and this DES (~1.2 g/cm³, see Table 1) can contribute to the inaccurate estimation of missing percentage using the Fee-Maroncelli method [100], and therefore the actual missing portion of the dynamics may be different from those indicated in Table 3. Temperature dependence of the measured solvation response function, $S(t)$, is shown in the lower panel of Fig. 8. Corresponding bi-exponential fit parameters are summarized in Table 3. Clearly, the solvation rate becomes faster with temperature, a trend already seen for the solute rotation in this DES. Fit parameters for $S(t)$ in Table 3 indicate that the measured solvation response is characterized by a dominating sub-nanosecond (~100–300 ps) component, followed by a slower long-time decay with timescale in nanoseconds (~1–3 ns).

We next explore the probable origin for these solvation timescales. Notice both the solvation timescales become shorter upon increasing the solution temperature; this suggests that solution structural relaxation is associated with these timescales [101]. In the absence of experimental dielectric relaxation data and simulations we speculate the following. The faster sub-nanosecond solvation component probably arises from both rotational relaxation of the DES particles and the faster part of the structural hydrogen bond relaxation involving the constituents [16]. Hydrodynamic centre-of-mass diffusion (translational

diffusion) cannot possibly contribute to this relatively faster timescale because translation diffusion timescales are in several to hundred nanoseconds and thus too slow (see Table S5). The slower multi nanosecond solvation timescale may have originated from the coupled orientational relaxation and the collective H-bond fluctuations involving the N—H group of urea acting as H-bond donor and C=O (of urea or/and betaine) as a H-bond acceptor [68]. This is explained schematically in Fig. S6 (Supporting material). However, this is speculative, and complete measurements of the solvation response and dielectric relaxation are needed for a better understanding of the polarization relaxation in this NADES.

It has already been mentioned that average solvation time ($\langle\tau_s\rangle$) becomes faster with temperature ($303 \leq T/K \leq 343$), reflecting the temperature dependence of the DES viscosity. In the upper panel of Fig. 9 we show in a double logarithmic fashion how $\langle\tau_s\rangle$ for C343 in this DES couples to the medium viscosity. Line going through the measured data depicts the following fractional viscosity dependence for average solvation times: $\langle\tau_s\rangle \propto (\eta/T)^p$ with $p \sim 0.7$. This fractional viscosity dependence of $\langle\tau_s\rangle$ indicates that viscous friction does not fully control the average rate of solvation of C343 in this medium. In other words, dynamically heterogeneous processes participate in and regulate the time dependent progress of solvation of excited C343 in this DES. Similar fractional viscosity dependence for $\langle\tau_s\rangle$ has already been observed for C153 solvation in several ionic acetamide DESs [14,23,72,92]. However, the value of the fraction power (p) in those ionic DESs was lower for C153 solvation than C153 rotation. This trend is therefore different from the results obtained for C343 solvation and rotation in this non-ionic DES.

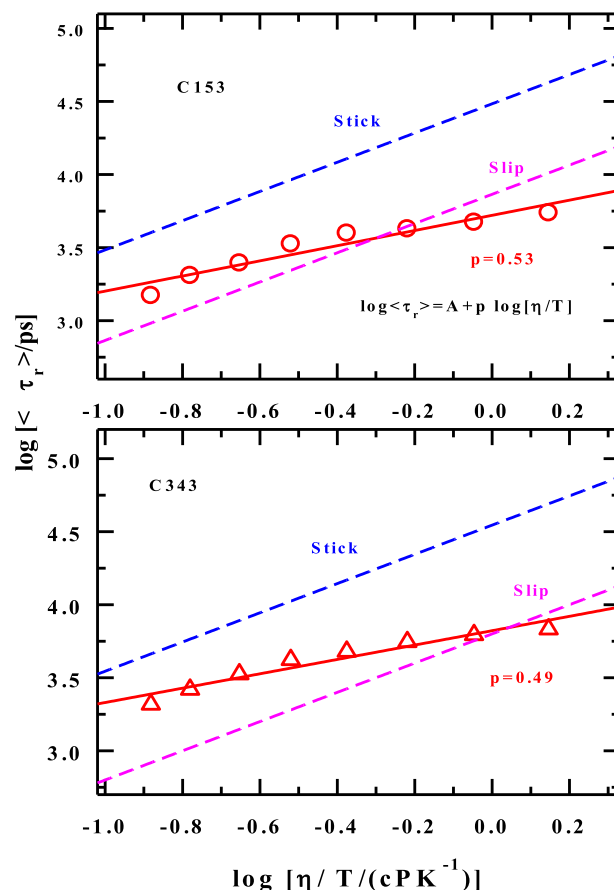


Fig. 6. Viscosity coupling of average rotation times ($\langle\tau_r\rangle$) for C153 and C343 in [Bet+Ure+Wat] DES. Temperature-dependent average rotational times are shown as a function of temperature-reduced viscosity (η/T) in a log-log fashion. Lines through the data represent fits to the following expression: $\log\langle\tau_r\rangle = A + p\log[\eta/T]$. Dashed lines represent the hydrodynamic (Stokes-Einstein-Debye) predictions.

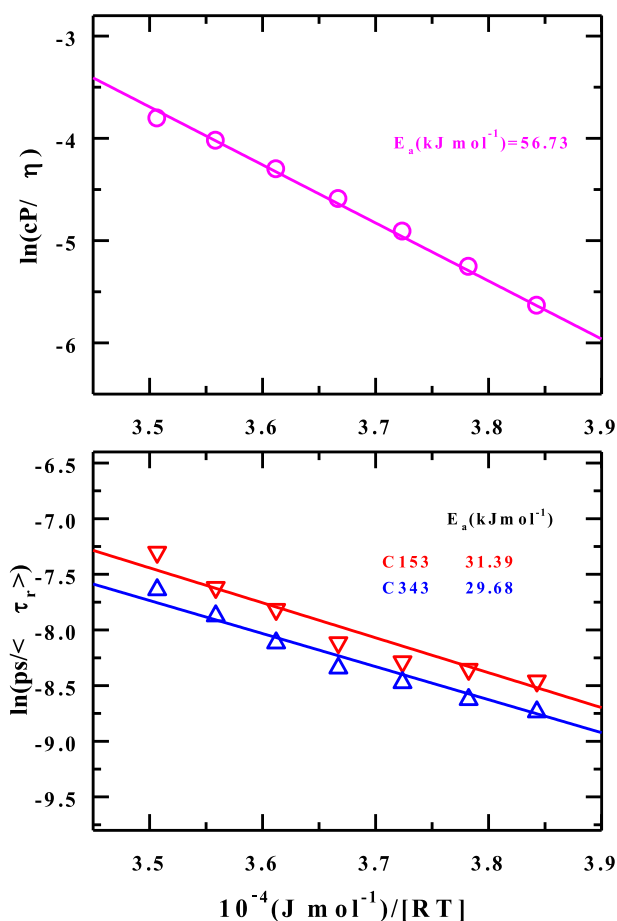


Fig. 7. Arrhenius plot for temperature dependent viscosity coefficients (η) for [Bet+Ure+Wat] DES, and average rotation times ($\langle\tau_r\rangle$) of two different solutes dissolved in it. Solid lines represent fits through the respective data sets. Note the different activation energies for η and $\langle\tau_r\rangle$. All representations are color coded.

Next, the coupling between solute rotation and solute solvation is explored in the lower panel of Fig. 9 where $\langle\tau_r\rangle$ is shown as a function of $\langle\tau_s\rangle$ in a log-log fashion. The line going through the data describes a fit to the relation, $\langle\tau_r\rangle \propto \langle\tau_s\rangle^m$ with $m \sim 0.8$. In the Fig. 9, we have also shown the same dependence, $\langle\tau_2\rangle \propto \langle\tau_s\rangle^m$ with $m \sim 0.72$, for the slower time constants (τ_2). The similar values for the exponent m reflect that the slower relaxations in solute solvation and solute rotation do not map the same medium friction. This result indicates that although medium friction impedes rates of both solvation and rotation of the solute, solute rotation explores medium friction differently than that by the solute solvation. In other words, solute solvation in this DES experiences frictional resistance more than solute rotation. If we recall that average solvation timescale is dominated by the centre-of-mass diffusion of the solvating particles [101], and the finding that translation diffusion in deeply supercooled liquids decouples from friction faster than rotation [104], then the present observation with C343 in this DES is new and contrasting. In addition, the value of m here is quite different from those found for C153 in ionic acetamide DES ($m \sim 1.09$) [23], in ionic liquids ($m \sim 1.03$) [94], and in vesicles ($m \sim 1.04$) [87]. We would also like to mention here that the ratio between these two average times, $\langle\tau_r\rangle/\langle\tau_s\rangle$, for C343 in this DES lies between ~ 5 – 7 (see Table 3) which is similar to the ratio found for C153 in ionic acetamide DES ($\langle\tau_r\rangle/\langle\tau_s\rangle \sim 5$) [23] but different from those found for C153 in dipolar ionic liquids ($\langle\tau_r\rangle/\langle\tau_s\rangle \sim 9$) [94], non-dipolar ionic liquids ($\langle\tau_r\rangle/\langle\tau_s\rangle \sim 2$ – 3) [105] and in vesicles ($\langle\tau_r\rangle/\langle\tau_s\rangle \sim 1$) [87]. These observations are, however, based on measurements with broad temporal resolution, which may alter if

reanalyzed after the measurements of full solvation response employing more sophisticated experimental technique and a neutral dipolar solvation probe, such as, C153. Measurements of dielectric relaxation and investigation of motional features via computer simulations using realistic potentials are required for understanding the microscopic mechanisms that govern the polarization relaxation dynamics measured in various time-domain and frequency-domain experiments. The focus of that study would be to explore the connection among the reorientational relaxations, H-bond fluctuations, and the dielectric relaxation of this complex ternary system, and then use that interrelationship to understand the frictional response for better designing of NADESs leading to smarter and large-scale chemical applications.

4. Conclusions

Here we have reported preparation of a new DES composed of betaine, urea and water ([Bet:Ure:Wat (11.7:12:1 weight ratio)]) and explored temperature dependent solute-centered relaxation dynamics employing time-resolved fluorescence measurements. This DES is stable at room temperature ($T_f = 279 \pm 2 \text{ K}$) with a measured $T_g \sim 218 \text{ K}$. Excitation wavelength dependent steady state fluorescence measurements using probes with widely different lifetimes suggest mild-spatial heterogeneity in the medium. Interestingly, time-resolved fluorescence anisotropy and Stokes shift measurements suggest strong fractional viscosity dependence for solute rotation and solvation times,

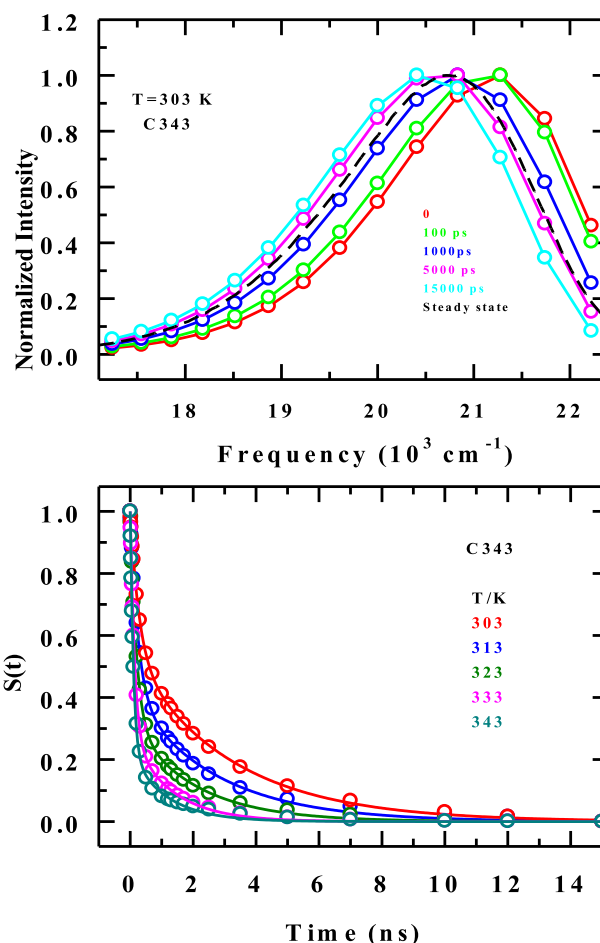


Fig. 8. Time-resolved emission spectra (TRES) (upper panel) at different time slices constructed from the measured intensity decays of C343 dissolved in [Bet+Ure+Wat] DES and the corresponding steady state emission spectrum. Temperature dependent solvation response functions, $S(t)$, of C343 in [Bet+Ure+Wat] DES are shown in the lower panel. Solid lines through the data represent bi-exponential fits to the $S(t)$. Fit parameters are shown in the Table 3.

Table 3

Bi-exponential fit parameters for solvation response functions, $S(t)$, obtained with C343 in [Bet+Ure+Wat] DES at various temperatures along with observed dynamic Stoke shift (ΔV_{obs}) magnitudes and missing percentages.

$T(K)$	a_1 (%)	τ_1 (ps)	a_2 (%)	τ_2 (ps)	$\langle \tau_s^{C343} \rangle^a$ (ps)	ΔV_{obs}^b (cm^{-1})	% missed	$\langle \tau_r \rangle / \langle \tau_s \rangle$
303	46	284	54	3178	1847	770	69	
313	58	250	42	2680	1271	830	70	4.9
323	68	197	32	2033	785	779	72	6.1
333	72	136	28	1291	459	705	75	7.3
343	78	102	22	1154	333	675	76	6.2

^a Average solvation time, $\langle \tau_s \rangle$, can be reproduced within $\pm 10\%$ uncertainty.

^b Uncertainty $\pm 150 \text{ cm}^{-1}$.

reflecting pronounced temporal heterogeneity. This is counter-intuitive in the sense that one often defines temporal heterogeneity in terms of relaxations whose rates are different for different micro-domains in a given macroscopic system. Interestingly, excitation wavelength dependence of emission energies for solute probes with differing lifetimes ($\sim 1\text{--}5 \text{ ns}$) suggest mild spatial medium heterogeneity. Both dynamic anisotropies and Stokes shift dynamics are strongly bi-exponential yet exhibit dissimilar coupling to medium viscosity, reflecting different frictional profiles probed by the solute rotation and the solute solvation.

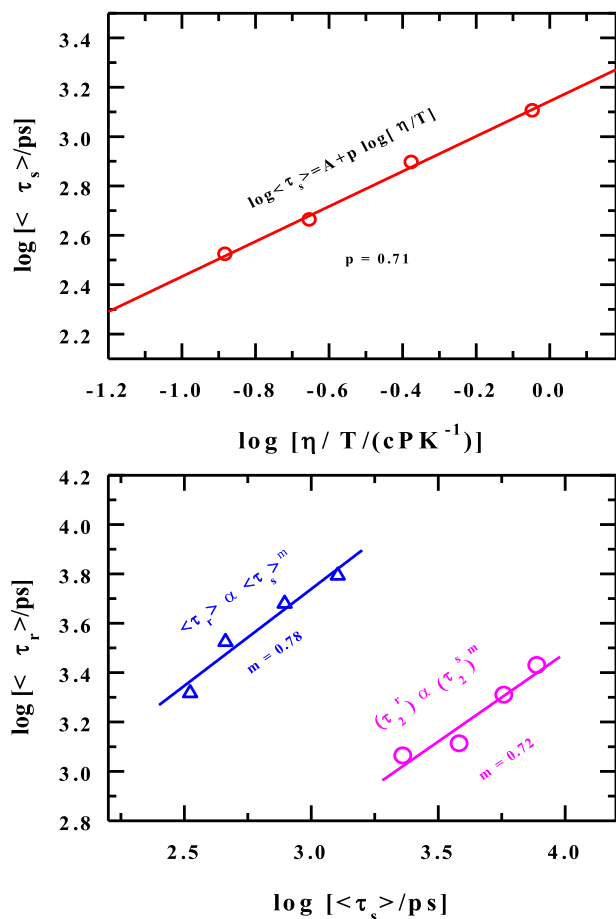


Fig. 9. Viscosity coupling of average solvation time ($\langle \tau_s \rangle$) of C343 in [Bet+Ure+Wat] DES (upper panel). Red solid line going through the data represents a linear fit. Lower panel shows correlation between $\langle \tau_r \rangle$ and $\langle \tau_s \rangle$ (also τ_2^r and τ_2^s) of C343 in [Bet+Ure+Wat] DES. Blue and pink lines denote linear fits and depict the inter-relationship between them.

Temperature dependent measurements indicate that the bi-exponential solvation response is characterized by time constants of $\sim 100\text{--}300 \text{ ps}$ and $\sim 1\text{--}3 \text{ ns}$, and a large missing portion. The biphasic solvation response is speculated to arise from structural H-bond relaxations, and restricted reorientations of H-bonded species coupled to collective H-bond relaxations. Further analyses suggest the friction that regulates the solute rotation is different from that controlling the solute solvation. This result is new, and contrasting. One needs to know here whether restricted reorientational relaxations coupled to intermolecular H-bonding relaxations contribute to time-dependent solvation through polarization relaxation at long-time. Complete measurements of solvation response employing sharper temporal resolution and megahertz-gigahertz dielectric relaxation in combination with computer simulations are required to explore and fully understand the relaxation dynamics in this and related [106] biodegradable deep eutectic solvents. In addition, the solution structural aspects need to be studied along the line of what has been done recently for several ionic and neutral DESs [77,107–109] for establishing the connection between temporal and spatial heterogeneities in these industrially important solvent systems.

In the earlier fluorescence studies of ionic deep eutectic solvents such as (alkylamide+electrolyte) [14], (acetamide+LiBr/LiNO₃) [15,92], (acetamide+sodium/potassium thiocyanates) [23] and (Choline Chloride + Urea) [72], signatures of both spatial and temporal heterogeneities were observed respectively via the excitation wavelength dependence of solute emission peak frequency and the fractional viscosity dependence of average rotation and solvation times. Interestingly, nonionic acetamide+urea DESs was found to be spatially and temporally homogeneous via steady state and time-resolved fluorescence measurements [21]. Addition of polyethylene glycol (PEG) in new acetamide+urea+PEG deep solvent was found to introduce temporal heterogeneity, although no evidence of spatial heterogeneity detected [68]. Similar results (as like acetamide+urea+PEG) also obtained for glucose [106] based DES and the present system. Therefore, we may state that heterogeneity in DESs does not arise solely from the deep depression of freezing points of the components; the nature of the constituents appears to play a critical role. It would be informative to study the low-frequency collective dynamics as done previously [69] because such studies can provide useful information on local interactions in this biodegradable solvent system.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgements

We thank an anonymous reviewer for a patient and careful reading of the manuscript, and valuable suggestions which helped improving

the quality of the manuscript. JM acknowledges the Department of Science and Technology, India for a fellowship under the INSPIRE scheme. ET acknowledges the SNBNCBS for a Visiting Research Associateship.

Appendix A. Supplementary data

See supplementary material for details of sample preparation, temperature dependent average fluorescence lifetimes of C153 and C343, representative parallel and perpendicular polarization intensity decays for C153 and C343 in this DES, magic angle intensity decays of C343 at the blue and red wavelengths, and temperature dependent hydrodynamic rotation and translational diffusion timescales of water, urea and betaine molecules in this DES. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2020.112451>.

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